

A 56-Gb/s Receiver Front-End with a CTLE and 1-Tap DFE in 20-nm CMOS

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Abstract

A 56-Gb/s receiver front-end suited for baud-rate clock recovery is demonstrated in 20-nm CMOS. Sharing the comparators for the data decision and phase detection minimizes the number of comparators in the front-end and reduces the power consumption. The front-end has a continuous-time linear equalizer followed by a 1-tap speculative decision-feedback equalizer. The front-end operates at 56Gb/s with a bit error rate of less than 10^{-12} with a 0.4UI margin in the bathtub curve. It occupies 0.27mm^2 and consumes 177mW of power from a 0.9-V supply.

Introduction

To meet ever-increasing demands for computing power in data centers, data rates over 50Gb/s/signal (e.g., OIF CEI-56G-VSR) will eventually be required in wireline chip-to-chip communications within and between servers. For receivers with such a high data rate, minimizing the number of front-end comparators and associated input-capacitance is crucial to extend the bandwidth of the continuous-time equalizer (CTLE) that feeds the incoming signal to the comparators. The loading of the clock signals to drive the comparators should also be minimized to keep the power consumption within a tight budget.

Baud-rate architectures [1] potentially reduce the comparator counts by halving the sampling phases, but this reduction is offset by the need for extra comparators to perform the phase detection from baud-rate samples. We propose a baud-rate clock-and-data-recovery scheme in which no extra comparators are needed for the phase detection other than the ones already implemented for speculative decision feedback equalizing (DFE). In this paper, we present the design and the measured results of the proposed front-end, which includes a CTLE, comparators, and a demultiplexer (DMX).

Receiver Architecture

We designed a receiver front-end for the proposed baud-rate scheme in which one of the two comparators assigned for 1-tap speculative DFE is used for the data decision depending on the previous bit's decision while the other is used for phase detection (Fig. 1). A pattern filter is used to select a bit usable for the phase detection. When, for example, a three-bit sequence '011' is detected, the center '1' bit is used for the phase detection, with the lower-threshold comparator output, DL, being used for the data decision and the higher-threshold comparator output, DH, for the phase detection. The CTLE should be adjusted so that the comparator assigned for the phase detection produces 0 and 1 at an equal probability when the decision timing is at the eye center.

In the proposed front-end, the incoming signal first goes through a T-coil-based matching network and enters into the CTLE (Fig. 2). The output of the CTLE is digitized by 4-way-interleaved comparator arrays driven by four-phase quarter-rate (i.e., 14-GHz) clock signals supplied from a

frequency divider. Each comparator array consists of two comparators used for both 1-tap speculative DFE and baud-rate phase detection. The resulting two four-bit data are demultiplexed by a 4-to-16 DMX, which is followed by the selectors for DFE data selection and data monitoring.

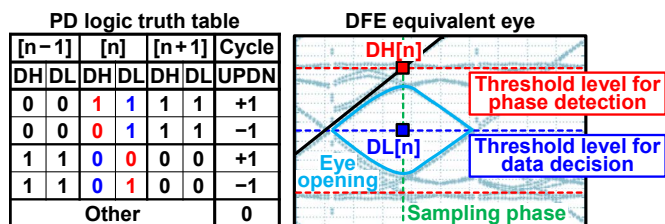


Fig. 1 Phase-detection scheme of the '011' bit sequence case.

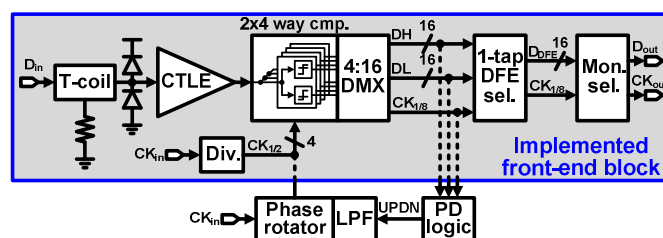


Fig. 2 Receiver architecture.

Circuit Implementation

The CTLE consists of a high-frequency equalizer that uses a source-degenerated CML stage and a low-frequency equalizer with a feedback loop that implements a low frequency pole and zero [2] (Fig. 3). The CTLE has a gain peak around 18GHz. Center-tapped multi-layer differential inductors are used to enhance the CTLE bandwidth with a small-area overhead. The inductor used in the CTLE is seven-turned and $10\mu\text{m} \times 20\mu\text{m}$ rectangular with an inductance value of 470pH.

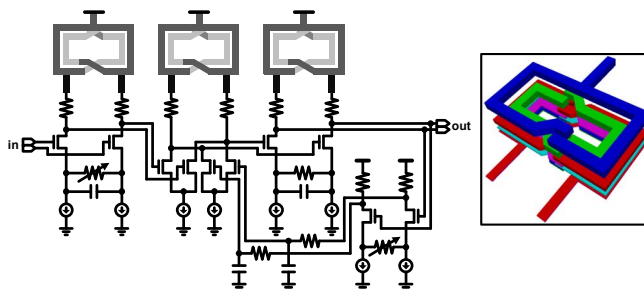
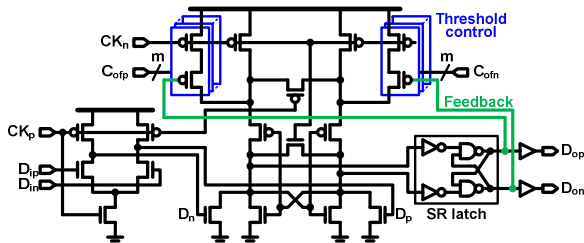


Fig. 3 CTLE with center-tapped multi-layer differential inductors.

The comparator is based on a double-tail sense amplifier followed by an SR latch [3] (Fig. 4). Split current-tail transistors in the regeneration stage controlled by a digital code inject current to the PMOS sources of the cross-coupled inverters and adjust the comparator threshold. Parasitic capacitance due to threshold control devices only occurs in the tail drain nodes, leading to negligible speed penalty. The threshold control is used for comparator's mismatch calibration and DFE-coefficient setting. Comparator history effect caused by incomplete reset is removed by adding a

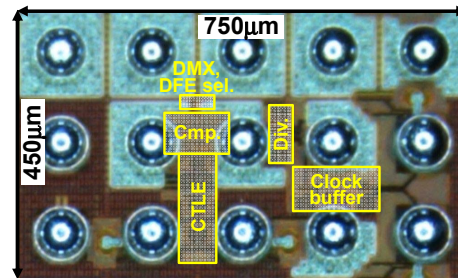
Although we use a speculative DFE scheme, it is still difficult to meet the 1UI limit of the decision feedback at the data rate of 56Gb/s. We thus adopt a look-ahead technique [4] with a modification for parallel-data processing in the DFE selector placed in the 3.5-GHz domain (Fig. 5). The DFE selector receives two pieces of 16-bit data, one from the higher- and the other from the lower-threshold comparators. The selector logic is divided into four identical blocks, each of which performs a selection based on one-, two-, or three-bit-earlier decisions among pre-calculated (i.e., look-ahead) signals. The gates in the critical path of the data selection are four selectors and one flip-flop, which are easily accommodated in the 3.5-GHz clock cycle.



The diagram illustrates the critical path of a 16-bit multiplier. It shows a series of logic blocks and flip-flops (FF) arranged in a pipeline. The inputs are DH[0], DL[0] and DH[1], DL[1]. The outputs are D_DFE[0] through D_DFE[15]. A red line traces the critical path, starting from the input of the first FF, passing through several logic blocks and FFs, and ending at the output of the last FF. The path is labeled 'Critical path' in red text.

Experimental Results

As a receiver front-end intended for a data rate over 50Gb/s, this work exhibited the highest level of integration with reasonable power consumption, timing margin, and feature set that can readily be used in a complete receiver [Table 1].



2x 28Gb/s
2ch. PPG
28GHz

2:1 MUX

2-tap FFE

56Gb/s

1-m cable

4-cm trace

DUT

Package

28 GHz

Dif. Amp.

Phase Shifter

BERT

3.5Gb/s

3.5GHz

Evaluation board

Input clock phase offset [UI]	Bit error rate
0.15	~0.02
0.20	~0.001
0.25	~10 ⁻¹²
0.65	~10 ⁻¹²
0.70	~10 ⁻⁶
0.75	~0.001
0.80	~0.02

Table 1 Performance summary.

Reference	[2]	[5]	This work
Data rate [Gb/s]	32	66	56
Equalizer	CTLE 2-tap DFE	3-tap DFE	2-tap FFE ^{*1} CTLE/1-tap DFE
Loss [dB]	40	—	23
Technology	28-nm CMOS	65-nm CMOS	20-nm CMOS
Supply [V]	0.9	1.2	0.9
Power [mW]	240	46 ^{*2}	177 ^{*3}

Conclusion

References

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- [4] K. K. Parhi, "Design of Multigigabit Multiplexer-Loop-Based Decision Feedback Equalizers," *IEEE Trans. on VLSI Systems*, vol. 13, no. 4, pp 489–493, Apr. 2005.
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